LOVELY PROFESSIONAL UNIVERSITY

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

FINAL INTERNSHIP REPORT

CLOUD INFRASTRUCTURE MONITORING, SECURITY AUDITING AND AUTOMATED ALERTING IN MULTI-CLOUD ENVIRONMENTS

An Internship Report Completed at WiSys, Riyadh, Saudi Arabia

Submitted in partial fulfillment of the requirements for the award of the degree of
BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
Specialization/Minor: Cloud Computing

Course Code: CSE443 | Term: 26271

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INTERNSHIP PERIOD: 14 June 2026 to 13 August 2026

LOVELY PROFESSIONAL UNIVERSITY
PHAGWARA, PUNJAB
AUGUST 2026

INTERNSHIP CERTIFICATE

This page is reserved for the official internship certificate issued by WiSys. The certificate should confirm that Aquib Jawaid Ansari, Registration Number 12508688, successfully completed industry training at WiSys, Riyadh, Saudi Arabia, from 14 June 2026 to 13 August 2026.

الشركة الوطنية لنظم المعلومات
AlWatania Information Systems

شهادة إتمام تدريب
Training Completion Certificate

Date: 13-Aug-2026				1448
Name :	Aquib Jawaid Ansari			
Passport	AH565204	AH565204		
Department :	Managed Services			
Training Period	Start Date	End Date		
	14-Jun-2026	13-Aug-2026	13-Aug-2026	14-Jun-2026

AL-Watania Information Systems Company

certifies that the mentioned Trainee has completed his Trining with the company. During the period shown in the above schedule

He proved to be hardworking, honest and punctual in his performance.
A high level of cooperation was exhibited in his Training at Wisys.

This certificate was issued upon his request without any responsibility to the company to the purpose that may serve him best.

*** (Any obliteration or alteration in the text of this certificate renders it cancelled).**

إدارة رأس المال البشري
Human Capital Management

Al Watania Information Systems Company Ltd.

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Field	Details
Industry Mentor	Emadeldin Taha Mahrous
Company Supervisor	Medhat
Organization	WiSys, Riyadh, Saudi Arabia

STUDENT DECLARATION

I hereby declare that this internship report titled "Cloud Infrastructure Monitoring, Security Auditing and Automated Alerting in Multi-Cloud Environments" is an authentic record of the work performed by me during my internship at WiSys, Riyadh, Saudi Arabia, from 14 June 2026 to 13 August 2026.

The report describes the technical activities, learning outcomes, troubleshooting work and academic project developed from the knowledge gained during the internship. No confidential company credentials, passwords, access keys, private customer information or restricted production information have been included in this report.

This report is submitted in partial fulfillment of the requirements of CSE443 at Lovely Professional University.

Date: 20 August 2026

Student Signature: ____Aquib Jawaid Ansari_____

Name: Aquib Jawaid Ansari
Registration Number: 12508688

ACKNOWLEDGEMENT

I sincerely thank WiSys, Riyadh, Saudi Arabia, for providing me with the opportunity to gain practical experience in cloud computing, infrastructure monitoring and IT operations.

I am especially thankful to my Industry Mentor, Emadeldin Taha Mahrous, and my Company Supervisor, Medhat, for their technical guidance, support and encouragement throughout the internship.

I would also like to thank the Cloud Operations and IT Infrastructure team for allowing me to work on practical tasks involving cloud monitoring, security auditing, automated alerting, troubleshooting and documentation.

I am also thankful to Lovely Professional University and the School of Computer Science and Engineering for providing the academic framework for this internship under course CSE443.

Finally, I would like to thank everyone who supported and guided me during the internship and helped me improve my technical and professional skills.

ABSTRACT

Modern organizations depend heavily on cloud infrastructure for hosting applications, databases, networks and business services. As cloud environments become larger and more distributed, continuous monitoring, security auditing and automated alerting become essential for maintaining system availability, reliability and security.

This report presents the work completed during an internship at WiSys, Riyadh, Saudi Arabia, from 14 June 2026 to 13 August 2026. The work focused on Google Cloud Platform (GCP) and Huawei Cloud, with practical exposure to cloud infrastructure monitoring, security auditing, alert configuration, troubleshooting and automation. The internship involved monitoring approximately 40 virtual machines and related cloud resources, with attention to VM lifecycle events, health-check failures, high disk utilization, security-group modifications, VPC deletion, backup failures, audit-log changes and monitoring-policy changes.

In GCP, Cloud Logging, Cloud Monitoring and Cloud Audit Logs were used to understand and monitor infrastructure operations. Log-based custom metrics were applied to important VM lifecycle events, and security findings were reviewed and documented in a structured report. Another major contribution was the improvement of alert emails by adding available incident details such as instance name, IP address, event type, user identity and source IP, enabling administrators to perform an initial assessment without immediately opening the cloud console.

Huawei Cloud exposure included Cloud Trace Service (CTS), Cloud Eye, Simple Message Notification (SMN), Elastic Cloud Server (ECS) and Virtual Private Cloud (VPC). Troubleshooting work covered SMTP sender validation, alert deduplication, stale alert records, JSON configuration portability and event simulation. The internship also included Microsoft Entra ID configuration for the WiSys AI Hub application and completion of an AWS certification.

After the internship ended and access to the company cloud environment was no longer available, a standalone Multi-Cloud Monitoring Simulation was developed for academic evaluation. The simulation demonstrates resources, infrastructure events, alert rules, alert processing, alert deduplication and customized notifications in a controlled, non-production environment.

Keywords: Cloud Monitoring, GCP, Huawei Cloud, Security Auditing, Automated Alerting, Cloud Audit Logs, Multi-Cloud, Microsoft Entra ID.

TABLE OF CONTENTS

No.	Chapter / Section	Page
1	Introduction	1
2	Organization Profile	2
3	Internship Role and Objectives	3
4	Technologies and Cloud Environment	4
5	Cloud Infrastructure Monitoring	5
6	GCP Monitoring and Security Auditing	6
7	Automated Alerting and Customized Notifications	7
8	Huawei Cloud Monitoring	8
9	Security Alert Configuration	9
10	Work Performed During the Internship	11
11	Major Troubleshooting Activities	13
12	WiSys AI Hub and Microsoft Entra ID	15
13	AWS Certification	16
14	Multi-Cloud Monitoring Simulation	17
15	Testing and Validation	21
16	Learning Outcomes	23
17	Challenges and Solutions	24
18	Conclusion	25

19	Future Scope	26
20	References	27
21	Appendices	28

LIST OF TABLES

1. Table 1. Student and Internship Details
2. Table 2. Technologies and Cloud Services Used
3. Table 3. Core Monitoring and Security Alerts
4. Table 4. Major Internship Contributions
5. Table 5. Simulated Cloud Resources
6. Table 6. Testing and Validation Results
7. Table 7. Challenge and Solution Log

LIST OF FIGURES

8. Figure 1. Conceptual Multi-Cloud Monitoring Architecture
9. Figure 2. AWS Certification Completed During Internship
10. Figure 3. Multi-Cloud Monitoring Simulation Dashboard
11. Figure 4. Simulated Infrastructure Alert
12. Figure 5. Customized Alert Notification

LIST OF ABBREVIATIONS

GCP	Google Cloud Platform
AWS	Amazon Web Services
VM	Virtual Machine
ECS	Elastic Cloud Server
CTS	Cloud Trace Service
SMN	Simple Message Notification
VPC	Virtual Private Cloud
IAM	Identity and Access Management
SMTP	Simple Mail Transfer Protocol
API	Application Programming Interface
JSON	JavaScript Object Notation
IP	Internet Protocol
SLA	Service Level Agreement
MTTR	Mean Time to Resolution
LPU	Lovely Professional University

1. INTRODUCTION

1.1 Background of the Internship

I completed my internship at WiSys, Riyadh, Saudi Arabia, from 14 June 2026 to 13 August 2026. The main focus of the internship was cloud infrastructure monitoring, security auditing, automated alerting and IT infrastructure support. During the internship, I worked with cloud resources hosted on Google Cloud Platform and Huawei Cloud. My responsibilities included monitoring infrastructure, configuring alerts, analysing logs, reviewing security findings, troubleshooting technical issues and preparing technical documentation.

Table 1. Student and Internship Details

Item	Details
Student Name	Aquib Jawaid Ansari
Registration Number	12508688
Programme	B.Tech Computer Science and Engineering
Specialization / Minor	Cloud Computing
Course Code / Term	CSE443 / 26271
Organization	WiSys, Riyadh, Saudi Arabia
Internship Period	14 June 2026 to 13 August 2026
Industry Mentor	Emadeldin Taha Mahrous
Company Supervisor	Medhat

1.2 Importance of Cloud Computing

Cloud computing enables organizations to use computing resources such as virtual machines, storage, networks and applications without depending only on physical infrastructure. Cloud platforms make it possible to create, modify and manage infrastructure quickly; however, larger environments require disciplined monitoring because a single unexpected change can affect applications and users.

1.3 Importance of Cloud Monitoring

Cloud monitoring helps administrators identify unexpected VM shutdowns, restarts and deletions; health-check failures; high disk utilization; network changes; backup failures; security changes and monitoring failures. Automated alerts allow the responsible team to receive important information quickly.

1.4 Importance of Security Auditing

Cloud audit logs record administrative and system activities. Depending on the event, the available information may include user identity, source IP, resource name, operation, project, event time and operation details. This information supports both monitoring and security investigation.

1.5 Academic Relevance

The internship was directly related to the B.Tech Computer Science and Engineering programme and Cloud Computing specialization. It enabled the practical application of cloud computing, computer networks, operating systems, cybersecurity, Python programming, database concepts, system administration, software engineering and cloud architecture.

2. ORGANIZATION PROFILE

2.1 WiSys

WiSys is a technology and IT services organization based in Riyadh, Saudi Arabia. The organization works in areas related to technology integration, cloud infrastructure, IT operations, cybersecurity and digital solutions. During the internship, the work was associated with cloud operations and IT infrastructure support.

2.2 Cloud Operations

Cloud operations involve managing and monitoring infrastructure to maintain service availability and security. A monitoring system helps an organization detect incidents, identify affected resources, understand what happened, notify administrators, reduce response time and maintain availability. These activities formed an important part of the internship.

3. INTERNSHIP ROLE AND OBJECTIVES

3.1 My Role

I worked as a Cloud / IT Infrastructure Trainee. The role combined infrastructure observation, alert configuration, troubleshooting, automation and documentation. Key activities included:

- Cloud infrastructure monitoring
- Alert configuration
- GCP log analysis
- GCP security findings review
- Security report preparation
- Huawei Cloud monitoring
- Audit-log analysis
- SMTP troubleshooting
- Python automation
- Database-related troubleshooting
- JSON configuration work
- Microsoft Entra ID configuration
- Technical documentation

3.2 Main Objectives

1. Understand enterprise cloud infrastructure.
2. Learn practical cloud monitoring.
3. Configure infrastructure alerts.
4. Understand cloud audit logs.
5. Identify and document security findings.
6. Improve automated email notifications.
7. Troubleshoot cloud-related problems.
8. Automate repetitive technical tasks.
9. Develop practical cloud engineering skills.

4. TECHNOLOGIES AND CLOUD ENVIRONMENT

The table below details the technologies, platforms, and services used during the internship training:

Table 2. Technologies and Cloud Services Used

Technology / Service	Purpose	Exposure
Google Cloud Platform	Cloud infrastructure	Hands-on
Compute Engine	Virtual machines	Hands-on
Cloud Logging	Log analysis	Hands-on
Cloud Monitoring	Monitoring and alerts	Hands-on
Cloud Audit Logs	Activity auditing	Hands-on
Huawei Cloud	Cloud infrastructure	Hands-on
Cloud Trace Service	Audit events	Hands-on
Cloud Eye	Monitoring and alarms	Hands-on
SMN	Notifications	Hands-on
Microsoft Entra ID	Application identity	Hands-on
Brevo SMTP	Email delivery	Hands-on
Python	Automation	Hands-on
Linux / Bash	Server administration	Practical
JSON	Configuration portability	Hands-on
AWS	Cloud knowledge	Certification

5. CLOUD INFRASTRUCTURE MONITORING

5.1 Multi-Cloud Monitoring

The internship involved monitoring activities across GCP and Huawei Cloud. Approximately 40 virtual machines and related resources were monitored across different cloud accounts and projects. The general monitoring process involved collecting information from cloud resources, analysing logs or metrics, applying monitoring rules, generating alerts and sending notifications to administrators.

5.2 VM Lifecycle Monitoring

Important VM lifecycle events included VM stopped, VM restarted and VM deleted. These events are significant because unexpected changes can cause service interruption or indicate an unauthorized action.

5.3 Health Monitoring

Health monitoring was used to identify whether important services or ports were responding correctly. A failed health check can indicate an application failure, network problem, port problem, VM problem or service failure.

5.4 Storage Monitoring

Disk utilization monitoring was configured around a 90% threshold. High disk usage can result in application failures, database write problems, logging failures and service instability.

5.5 Network Monitoring

Network-related changes were monitored, including security-group modifications and VPC deletion. These events may affect both security and availability.

6. GCP MONITORING AND SECURITY AUDITING

6.1 GCP Monitoring

The main GCP services used were Compute Engine, Cloud Logging, Cloud Monitoring and Cloud Audit Logs. One key task was monitoring VM lifecycle operations.

6.2 Log-Based Custom Metrics

Some infrastructure operations are better identified through audit logs rather than normal resource metrics. Log filters were used to identify important VM lifecycle events and convert relevant log events into monitoring signals, allowing infrastructure events to act as inputs for monitoring and alert generation.

6.3 Event Information

Depending on the event, audit information could include principal email, source IP, resource name, operation, project, event time and event details. This information was useful for generating customized alert notifications.

6.4 GCP Security Findings

Security findings in GCP were reviewed and organized into a security findings report. The report recorded the finding type, affected resource, security importance, available evidence, recommended action and status.

6.5 Security Findings Reporting Process

The process involved reviewing a security finding, identifying the affected resource, analysing the potential risk, documenting the issue and recommending an appropriate action. This work provided practical exposure to cloud security monitoring and security documentation.

6.6 Importance of the Security Review

Security findings can reveal configuration problems or potential risks. Reviewing and documenting findings helps administrators understand what the problem is, which resource is affected, why it is important and what action should be taken.

7. AUTOMATED ALERTING AND CUSTOMIZED NOTIFICATIONS

7.1 Basic Alerting Problem

A practical issue with basic alert notifications is that the email may not contain enough information for an administrator to immediately understand an incident. Searching the cloud console for an affected resource can increase investigation time.

7.2 Custom Alert Messages

Alert messages were improved by including useful incident information directly in the email. Depending on the event, the message could contain instance name, instance ID, IP address, event type, user or principal, source IP, project, resource, event time, severity and a short description.

7.3 Sample Custom Alert Structure:

Field	Sample Value
ALERT TYPE	VM INSTANCE STOPPED
CLOUD PROVIDER	GCP
INSTANCE	gcp-vm-prod-01
IP ADDRESS	192.168.x.x
USER	administrator
SOURCE IP	10.x.x.x
SEVERITY	CRITICAL
EVENT TIME	[EVENT TIME]
DESCRIPTION	The monitored virtual machine was stopped.
RECOMMENDED ACTION	Verify whether the operation was authorized and check the application status.

7.4 Advantage of Custom Alert Emails

Customized messages allow an administrator to understand the basic incident directly from the email: what happened, which instance was affected, who performed the action, what source IP was involved and when the event occurred. This reduces the need to immediately open multiple cloud-console pages.

7.5 Notification Process

The notification process involved detecting a cloud event, processing the corresponding log or metric, evaluating the alert rule, generating a customized message, sending it through the SMTP relay and delivering the notification to the administrator.

8. HUAWEI CLOUD MONITORING

8.1 Huawei Cloud

Huawei Cloud monitoring work involved Cloud Trace Service (CTS), Cloud Eye, Simple Message Notification (SMN), Elastic Cloud Server (ECS) and Virtual Private Cloud (VPC).

8.2 Cloud Trace Service

Cloud Trace Service records important operations performed on Huawei Cloud resources. It was used to understand administrative events and resource changes, with event details helping identify the resource, operation, user, time and result.

8.3 Cloud Eye

Cloud Eye was used for monitoring and alarm management. Work included alarm configuration and understanding the relationship between events, monitoring rules and notifications.

8.4 Simple Message Notification

SMN was used to route notifications through topics. The flow consisted of event detection, rule evaluation, Cloud Eye alarm generation and notification forwarding through an SMN topic.

9. SECURITY ALERT CONFIGURATION

The table below details the 11 monitored alerts configured during my internship:

Table 3. Core Monitoring and Security Alerts

Category	Alert	Severity	Main Risk
Compute	VM Instance Deleted	Critical	Loss of resource or system state
Compute	VM Instance Stopped	Critical	Application downtime
Compute	VM Instance Restarted	Critical	Unexpected service interruption
Compute	VM Health Check Failed	Critical	Service unavailable
Storage	Disk Utilization ≥ 90%	Critical	Storage exhaustion
Network	Security Group Modified	Critical	Possible security exposure
Network	VPC Deleted	Critical	Network disruption
Backup	Backup Failed	Critical	Data-loss risk
Security	Audit Log Disabled	Critical	Loss of audit visibility
Monitoring	Alarm Policy Deleted	Critical	Monitoring gap
Monitoring	Alarm Policy Disabled	Warning	Resource may become unmonitored

9.1 VM Instance Deleted

A VM deletion event can result in loss of the VM and its system state; it is therefore an important event to monitor.

9.2 VM Instance Stopped

An unexpected VM stop can cause application downtime. The alert allows administrators to identify the event quickly and investigate the reason.

9.3 VM Instance Restarted

An unexpected restart can indicate system failure, a software issue, an administrative action or an infrastructure problem.

9.4 VM Health Check Failed

A failed health check indicates that a service or system is not responding correctly.

9.5 Disk Utilization ≥ 90%

High disk usage can cause application and database problems; the 90% threshold was treated as an important alert condition.

9.6 Security Group Modified

Security-group modifications can change network access. An unexpected change can create a security risk.

9.7 VPC Deleted

VPC deletion can affect connected resources and network connectivity, and should be detected immediately.

9.8 Backup Failed

Backup failure increases the risk of data loss. Monitoring backup failures allows administrators to take corrective action.

9.9 Audit Log Disabled

Audit logs are important for security investigations. Disabling logging reduces visibility into administrative actions.

9.10 Alarm Policy Deleted or Disabled

Deleting or disabling monitoring policies can create monitoring gaps. These changes should therefore be detected and reviewed.

10. WORK PERFORMED DURING THE INTERNSHIP

The table below summarizes my specific engineering and operational tasks during the training:

Table 4. Major Internship Contributions

Area	Contribution
Multi-Cloud Monitoring	Configured and verified monitoring for approximately 40 virtual machines and related resources.
GCP Custom Metrics	Configured log filters to track VM lifecycle operations.
GCP Security	Reviewed security findings and prepared a security findings report.
GCP Alerts	Configured and tested monitoring alerts.
Custom Email Alerts	Added instance, IP, user, source and event information to notification messages.
Huawei Cloud	Worked with CTS, Cloud Eye and SMN.
SMTP	Troubleshot email delivery and sender validation problems.
Alert Database	Identified and resolved a backlog of 55 stale OPEN alerts.
JSON Configuration	Worked on reusable alarm configurations and project-specific parameters.
Event Simulator	Improved the simulator to use valid resource information.
WiSys AI Hub	Configured Microsoft Entra ID application registration and related settings.
Documentation	Prepared technical documentation and troubleshooting information.
AWS	Completed an AWS certification during the internship.

10.1 Multi-Cloud Monitoring Setup

Monitoring activities for approximately 40 virtual machines and related cloud resources were configured and verified to detect important infrastructure events and notify administrators.

10.2 GCP Custom Metrics

Log filters on GCP Audit Logs were configured to identify VM lifecycle operations and convert relevant log events into monitoring signals.

10.3 GCP Security Findings

GCP security findings were reviewed and documented in a security findings report, providing practical experience in identifying, analysing and documenting cloud security issues.

10.4 Custom Alert Emails

Alert messages were customized to include useful incident information, such as instance details, IP address, user, source IP and event information.

10.5 Huawei Cloud Monitoring

CTS, Cloud Eye and SMN were used to understand event monitoring and notification across the Huawei Cloud monitoring pipeline.

10.6 Technical Documentation

Documentation was prepared for configurations, troubleshooting procedures and monitoring workflows, helping technical teams understand system configuration and common problem resolution.

11. MAJOR TROUBLESHOOTING ACTIVITIES

11.1 SMTP Relay Sender Validation Problem

During testing, SMTP alert emails were not delivered correctly because the SMTP service rejected the configured sender identity. Configuration and logs were investigated, the sender validation problem was identified and the email configuration was corrected. Email delivery was restored after correction.

11.2 Alert Deduplication Queue Problem

The alerting system used deduplication logic to prevent repeated notifications for the same active alert. During testing, new alert emails were not generated. Database checks identified 55 stale OPEN alerts, which prevented new events from creating notifications. A Python cleanup script was created and executed to change stale alerts to RESOLVED. The stale backlog was removed, new alerts could be generated, notification testing worked correctly and the dashboard compliance score returned to 100/100.

11.3 JSON Configuration Import Problem

Exported JSON alarm configurations failed to import in some cases because they included project-specific identifiers. These values were parameterized so configurations could be reused with different project values, improving portability.

11.4 Event Simulator Problem

The event simulator initially used dummy resource IDs, meaning simulated events did not always update the correct dashboard resources. It was improved to use valid resource information from the database, making testing more realistic and associating simulated events with represented resources.

12. WISYS AI HUB AND MICROSOFT ENTRA ID

12.1 WiSys AI Hub

Identity configuration work was completed for the WiSys AI Hub application. The application required integration with Microsoft Entra ID for authentication.

12.2 Application Registration

Specific configurations included:

- Application registration in the Entra portal
- Client credential (secret) generation
- Redirect URI configuration
- API permission configuration
- Requesting User.Read permission
- Acquiring Administrator consent

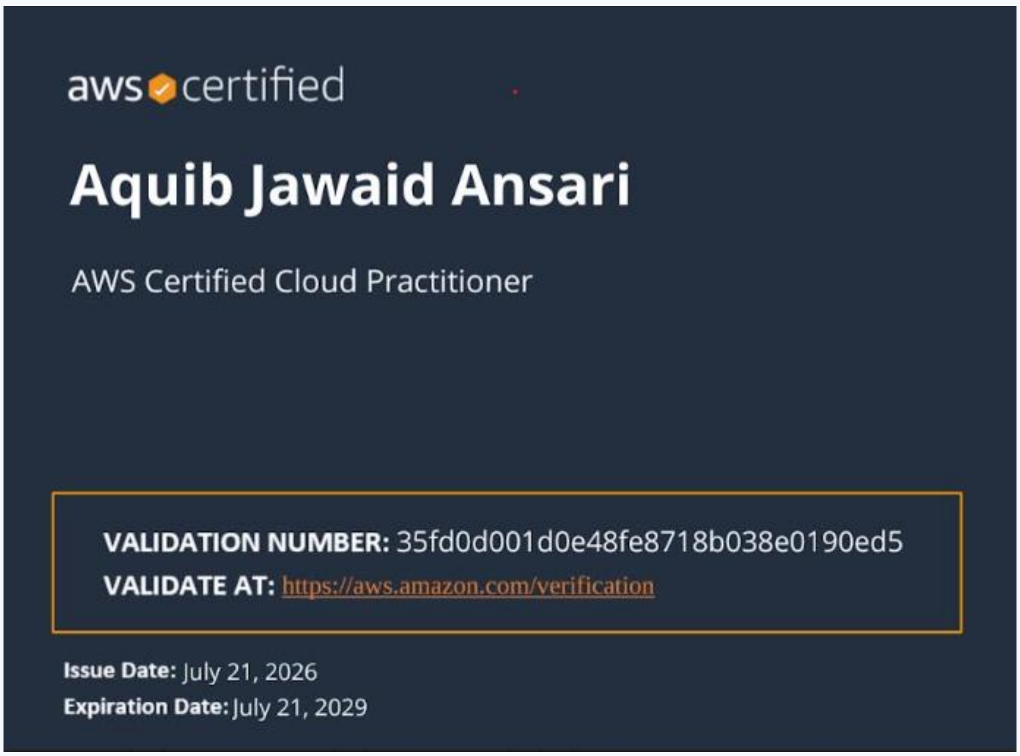
12.3 Importance of the Configuration

Microsoft Entra ID provides a centralized identity system for applications. The configuration enabled the WiSys AI Hub application to use standard authentication and authorization mechanisms, providing practical experience with application identity and access management.

13. AWS CERTIFICATION

13.1 Certification Completed During the Internship

An AWS certification was completed during the internship. It expanded cloud knowledge beyond the GCP and Huawei Cloud environments used in day-to-day internship work and strengthened understanding of cloud computing, AWS services, cloud infrastructure, cloud security, cloud architecture and cloud operations.



AWS Certification Completed During Internship

14. MULTI-CLOUD MONITORING SIMULATION

14.1 Project Background

After the internship ended, access to the company cloud environment was no longer available. Therefore, a standalone Multi-Cloud Monitoring Simulation was developed for academic evaluation and demonstration. The simulation is based on monitoring requirements and concepts learned during the internship, and it does not use WiSys credentials or company production infrastructure.

14.2 Project Objective

The objective is to demonstrate how resources from multiple cloud providers can be represented and monitored using a common monitoring system. The simulation covers GCP and Huawei Cloud resources, VM lifecycle events, security events, storage alerts, network alerts, backup alerts, monitoring alerts and automated notifications.

14.3 Problem Statement

In a multi-cloud environment, resources are distributed across providers. Monitoring each provider separately can make infrastructure management more difficult. The simulation provides a common monitoring environment in which resources and events from different providers can be represented together.

14.5 Resource Manager

The Resource Manager stores simulated resource information, including provider, resource ID, resource name, resource type, IP address, status and environment.

14.6 Event Generator

The Event Generator creates simulated infrastructure events such as VM stopped, VM restarted, VM deleted, health check failed, disk utilization high, security group modified, VPC deleted, backup failed, audit logging disabled and alarm policy disabled.

14.7 Monitoring Engine

The Monitoring Engine checks every generated event against configured alert rules. For example, when a VM stop event is generated, the engine identifies the VM Instance Stopped rule, assigns the appropriate severity and creates an alert.

14.8 Alert Database

The alert database stores generated alerts and their status. Important states include OPEN and

RESOLVED. It also supports alert deduplication so repeated events do not create unnecessary notifications.

14.9 Notification Generator

The Notification Generator creates detailed messages containing available event information such as cloud provider, instance name, instance ID, IP address, event, user, source IP, time, severity and description.

Table 5. Simulated Cloud Resources

Provider	Resource Name	Type	Status
GCP	gcp-vm-prod-01	Compute VM	RUNNING
GCP	gcp-vm-db-01	Compute VM	RUNNING
Huawei Cloud	huawei-ecs-prod-01	ECS	RUNNING
Huawei Cloud	huawei-ecs-db-01	ECS	RUNNING
GCP	gcp-vpc-prod	VPC	ACTIVE
Huawei Cloud	huawei-vpc-prod	VPC	ACTIVE

Note: These are simulated resources created for academic demonstration and are not company production resources.

14.10 Alert Deduplication

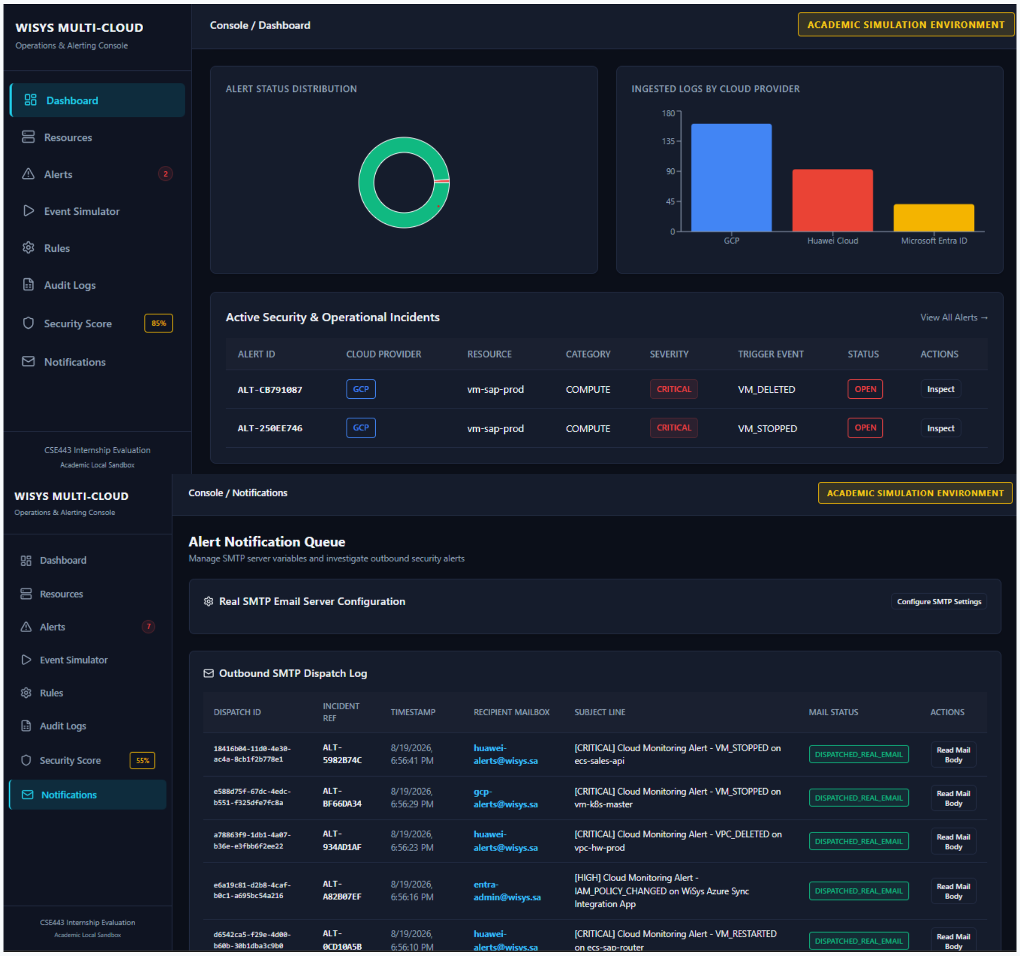
If the same resource repeatedly generates the same alert while a previous alert remains OPEN, the system can avoid creating unnecessary duplicate alerts. When the issue is resolved, the alert is changed to RESOLVED, allowing a new event to create a new alert. This concept reflects the alert troubleshooting work completed during the internship.

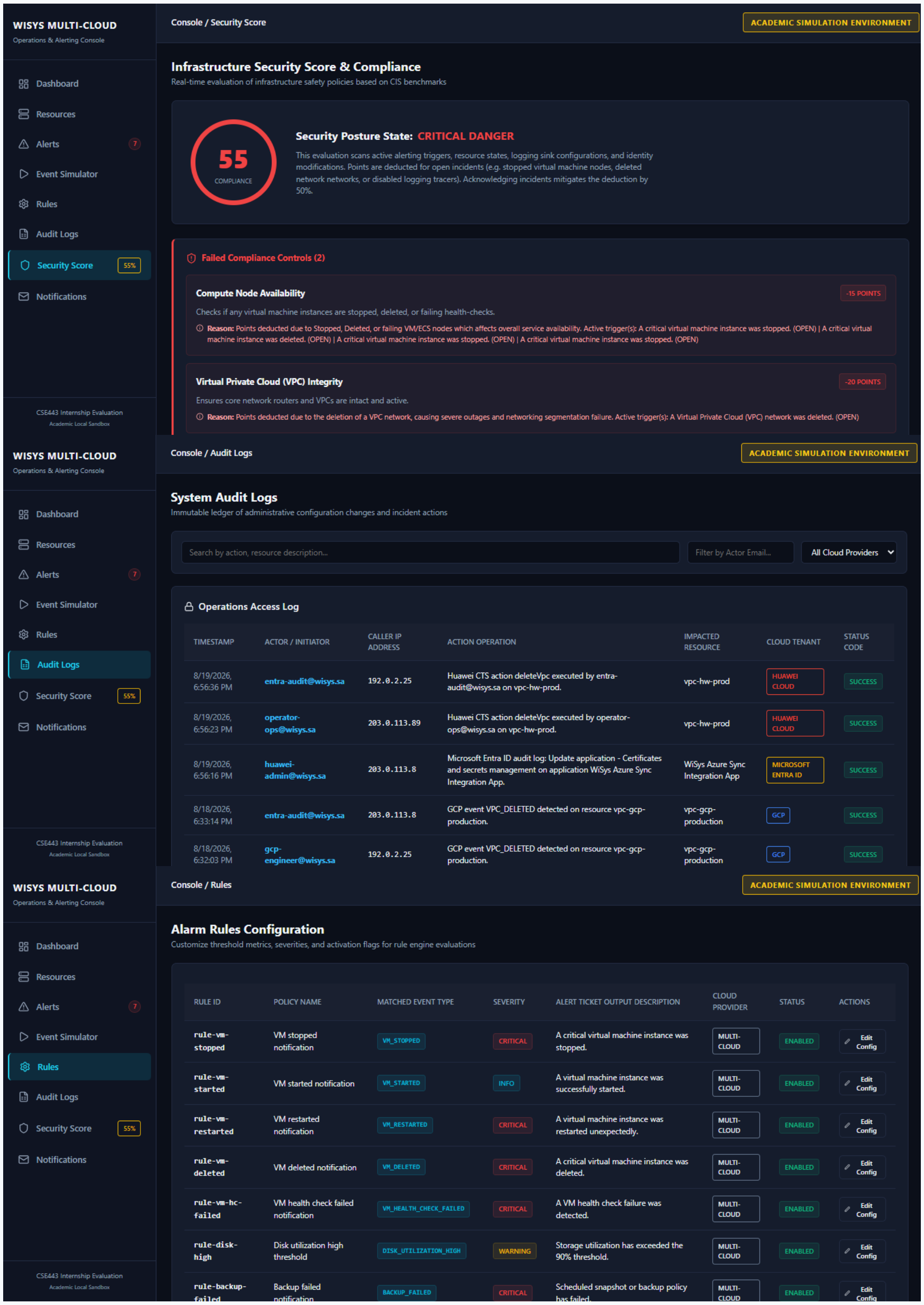
14.11 Security Monitoring

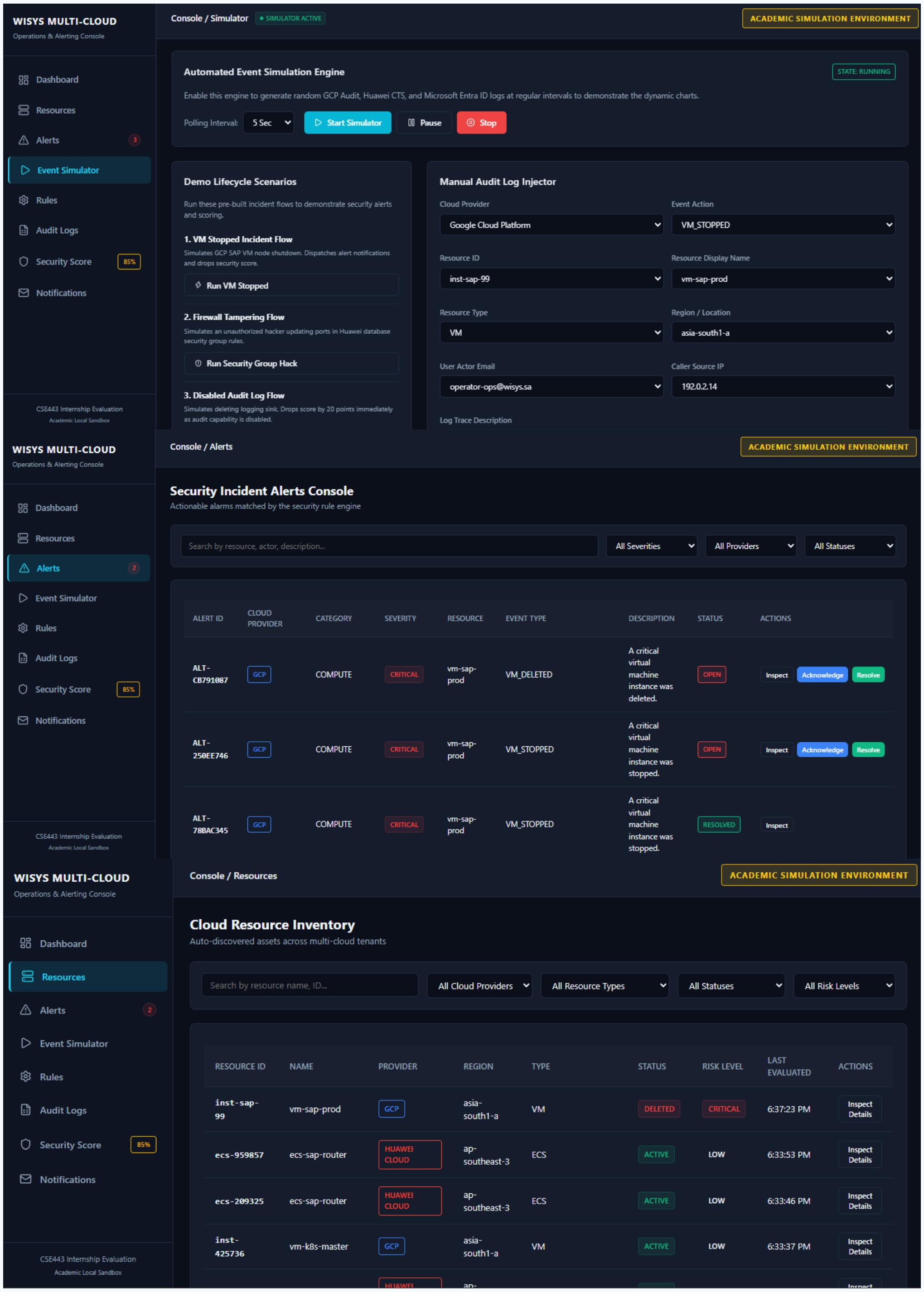
The simulation includes security-related alerts such as security group modification, audit logging disabled, VPC deletion and unauthorized-looking VM lifecycle events. Administrators can review the events through the dashboard and notification system.

14.12 Multi-Cloud Dashboard

The dashboard provides a single view of simulated resources and alerts. It can display total resources, running resources, stopped resources, open alerts, critical alerts, resolved alerts, cloud provider and resource status.




WISYS MULTI-CLOUD
Operations & Alerting Console
Dashboard
Resources
Alerts
7
Event Simulator
Rules
Audit Logs
Security Score
55%
Notifications
CSE443 Internship Evaluation
Academic Local Sandbox
Console / Security Score
ACADEMIC SIMULATION ENVIRONMENT
Infrastructure Security Score & Compliance
Real-time evaluation of infrastructure safety policies based on CIS benchmarks
55
COMPLIANCE
Security Posture State: CRITICAL DANGER
This evaluation scans active alerting triggers, resource states, logging sink configurations, and identity modifications. Points are deducted for open incidents (e.g. stopped virtual machine nodes, deleted network networks, or disabled logging tracers). Acknowledging incidents mitigates the deduction by 50%.
Failed Compliance Controls (2)
Compute Node Availability
-15 POINTS
Checks if any virtual machine instances are stopped, deleted, or failing health-checks.
Reason: Points deducted due to Stopped, Deleted, or failing VM/ECS nodes which affects overall service availability. Active trigger(s): A critical virtual machine instance was stopped. (OPEN) | A critical virtual machine instance was deleted. (OPEN) | A critical virtual machine instance was stopped. (OPEN) | A critical virtual machine instance was stopped. (OPEN)
Virtual Private Cloud (VPC) Integrity
-20 POINTS
Ensures core network routers and VPCs are intact and active.
Reason: Points deducted due to the deletion of a VPC network, causing severe outages and networking segmentation failure. Active trigger(s): A Virtual Private Cloud (VPC) network was deleted. (OPEN)
Console / Audit Logs
System Audit Logs
Immutable ledger of administrative configuration changes and incident actions
Search by action, resource description...
Filter by Actor Email...
All Cloud Providers
Operations Access Log
TIMESTAMP
ACTOR / INITIATOR
CALLER IP ADDRESS
ACTION OPERATION
IMPACTED RESOURCE
CLOUD TENANT
STATUS CODE
8/19/2026, 6:56:36 PM
entra-audit@wisys.sa
192.0.2.25
Huawei CTS action deleteVpc executed by entra-audit@wisys.sa on vpc-hw-prod.
vpc-hw-prod
HUAWEI CLOUD
SUCCESS
8/19/2026, 6:56:23 PM
operator-ops@wisys.sa
203.0.113.89
Huawei CTS action deleteVpc executed by operator-ops@wisys.sa on vpc-hw-prod.
vpc-hw-prod
HUAWEI CLOUD
SUCCESS
8/19/2026, 6:56:16 PM
huawei-admin@wisys.sa
203.0.113.8
Microsoft Entra ID audit log: Update application - Certificates and secrets management on application WiSys Azure Sync Integration App.
WiSys Azure Sync Integration App
MICROSOFT ENTRA ID
SUCCESS
8/18/2026, 6:33:14 PM
entra-audit@wisys.sa
203.0.113.8
GCP event VPC_DELETED detected on resource vpc-gcp-production.
vpc-gcp-production
GCP
SUCCESS
8/18/2026, 6:32:03 PM
gcp-engineer@wisys.sa
192.0.2.25
GCP event VPC_DELETED detected on resource vpc-gcp-production.
vpc-gcp-production
GCP
SUCCESS
Console / Rules
Alarm Rules Configuration
Customize threshold metrics, severities, and activation flags for rule engine evaluations
RULE ID
POLICY NAME
MATCHED EVENT TYPE
SEVERITY
ALERT TICKET OUTPUT DESCRIPTION
CLOUD PROVIDER
STATUS
ACTIONS
rule-vm-stopped
VM stopped notification
VM_STOPPED
CRITICAL
A critical virtual machine instance was stopped.
MULTI-CLOUD
ENABLED
Edit Config
rule-vm-started
VM started notification
VM_STARTED
INFO
A virtual machine instance was successfully started.
MULTI-CLOUD
ENABLED
Edit Config
rule-vm-restarted
VM restarted notification
VM_RESTARTED
CRITICAL
A virtual machine instance was restarted unexpectedly.
MULTI-CLOUD
ENABLED
Edit Config
rule-vm-deleted
VM deleted notification
VM_DELETED
CRITICAL
A critical virtual machine instance was deleted.
MULTI-CLOUD
ENABLED
Edit Config
rule-vm-hc-failed
VM health check failed notification
VM_HEALTH_CHECK_FAILED
CRITICAL
A VM health check failure was detected.
MULTI-CLOUD
ENABLED
Edit Config
rule-disk-high
Disk utilization high threshold
DISK_UTILIZATION_HIGH
WARNING
Storage utilization has exceeded the 90% threshold.
MULTI-CLOUD
ENABLED
Edit Config
rule-backup-failed
Backup failed notification
BACKUP_FAILED
CRITICAL
Scheduled snapshot or backup policy has failed.
MULTI-CLOUD
ENABLED
Edit Config



Multi-Cloud Monitoring Simulation Dashboard

14.13 Simulation Benefits

- Monitor different cloud providers through one conceptual interface.
- Standardize alert rules and severities.
- Display resources in one interface.
- Generate detailed alerts and customized notifications.
- Track alert status and reduce duplicate notifications.
- Support security monitoring and a basis for future automation.

15. TESTING AND VALIDATION

15.1 Testing Approach

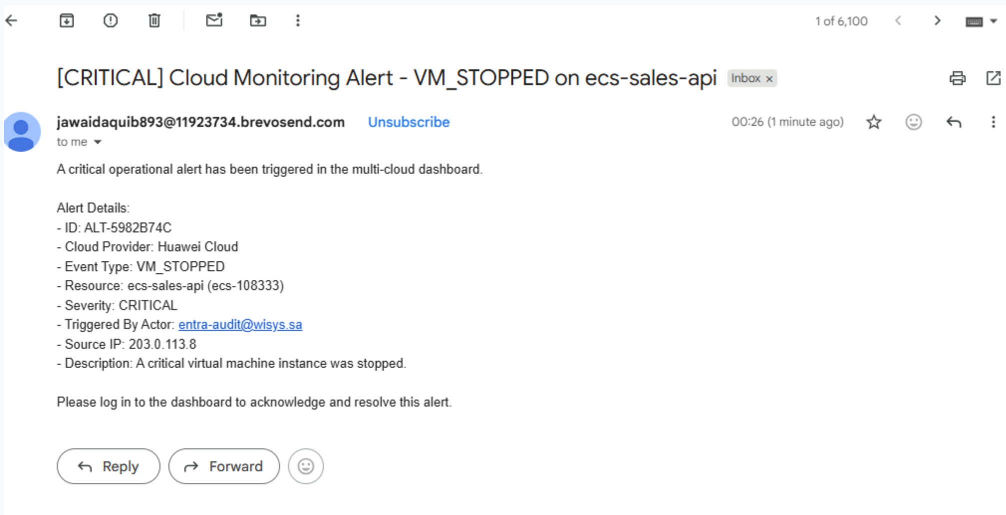
The monitoring simulation was tested by generating different infrastructure events and checking whether the correct alert was created. The tests verified event detection, alert type, severity, resource, status, notification information and deduplication behaviour.

Table 6. Testing and Validation Results

Test Event	Expected Result	Status
VM Stop	VM Stopped alert	Passed
VM Restart	VM Restarted alert	Passed
VM Delete	VM Deleted alert	Passed
Health Check Failure	Health alert	Passed
High Disk Usage	Storage alert	Passed
Security Group Change	Security alert	Passed
VPC Deletion	Network alert	Passed
Backup Failure	Backup alert	Passed
Audit Log Disabled	Security alert	Passed
Alarm Policy Deleted	Monitoring alert	Passed
Alarm Policy Disabled	Monitoring warning	Passed

15.2 Custom Email Test

The custom notification was tested to verify inclusion of important fields such as event, provider, resource, IP, user, source IP, severity and time.



Customized Alert Notification (Email)

16. LEARNING OUTCOMES

16.1 Cloud Observability & Competencies

- Cloud Computing: Practical understanding of cloud infrastructure, managing virtual machines, networks, and services.
- Cloud Monitoring: Setting up monitoring rules, metrics, and alerting workflows.
- Security Auditing: Utilizing cloud audit logs to track resource modifications.
- Alert Automation: Configuring notification topics, subscriptions, and SMTP relays.
- Troubleshooting: Solving sender validation rejections, clearing DB alert backlogs, and parameters portability.
- Identity Management: Managing application registrations, client secrets, and permissions in Microsoft Entra ID.

16.2 Professional Skills

The internship improved technical communication, ticket workflows, problem-solving, SLA compliance, and collaboration within a professional IT operations team.

17. CHALLENGES AND SOLUTIONS

The table below details the core technical challenges and applied solutions during the internship:

Table 7. Challenge and Solution Log

Challenge	Solution	Result
SMTP sender rejection	Corrected sender configuration to verified email address	Email delivery restored
55 stale OPEN alerts blocking queue	Ran Python database cleanup script to transition states to RESOLVED	New alerts processed correctly
JSON import compatibility errors	Parameterized environment-specific UUIDs and ARNs	Rules standardized for multi-project reuse
Dummy simulator resources	Refactored generator loop to target real database resource entities	Simulated events updated actual resources

18. CONCLUSION

The internship at WiSys, Riyadh, provided valuable practical experience in cloud infrastructure, monitoring, security auditing and IT operations. Work with GCP and Huawei Cloud provided hands-on exposure to cloud monitoring, audit logs, custom metrics, security findings, alert configuration and automated notifications.

Improving alert emails with instance name, IP address, event type, user and source IP made it possible for administrators to perform an initial analysis directly from an email. GCP security findings work provided experience in cloud security assessment and reporting. Troubleshooting tasks involving SMTP delivery, alert deduplication, database records, JSON configurations and event simulation improved debugging and automation skills.

Microsoft Entra ID configuration for WiSys AI Hub provided experience with application identity, while AWS certification further expanded cloud knowledge. Finally, the Multi-Cloud Monitoring Simulation converted practical concepts learned at WiSys into an academic project that can be demonstrated without access to company cloud infrastructure.

Overall, the internship helped bridge the gap between academic learning and real-world cloud infrastructure operations and provided a strong foundation for further work in Cloud Engineering, DevOps, Infrastructure Engineering and Cloud Security.

19. FUTURE SCOPE

1. Centralized multi-cloud log aggregation to view alerts from all cloud providers in a single console.
2. Automating monitoring deployments using Infrastructure as Code (IaC) tools like Terraform.
3. Implementing machine learning models to detect anomalies and unusual patterns in resource performance.
4. Creating automated remediation playbooks to resolve incidents (such as restarting crashed ports) automatically.
5. Integrating monitoring alerts with IT Service Management (ITSM) ticketing systems.

Note: Company-internal material is referenced only at a high level. No confidential credentials, customer data or restricted production information is included in this report.

21. APPENDICES

APPENDIX A – CONCEPTUAL ARCHITECTURE

[GCP & Huawei Cloud Resources] ➔ [Resource Manager] ➔ [Event Generator] ➔ [Monitoring Engine & Rules] ➔ [Alert DB & Deduplication] ➔ [Notification Engine & Dashboard] ➔ [Administrator]

APPENDIX B – CUSTOM CONFIGURATION FORMATS

Example 1: Parameterized JSON Config Template for SMTP Service (smtp_config.json)

```
{
  "smtp_server": "smtp-relay.brevo.com",
  "smtp_port": 587,
  "smtp_username": "b5f116001@smtp-brevo.com",
  "smtp_password": "[REDACTED_PASSWORD_KEY]",
  "alert_recipient": "jawaidaquib893@gmail.com"
}
```

Example 2: Sanitized GCP Custom Log Metric Filter

```
resource.type="gce_instance"
protoPayload.methodName="v1.compute.instances.stop"
protoPayload.authenticationInfo.principalEmail="*@wisys.sa"
```